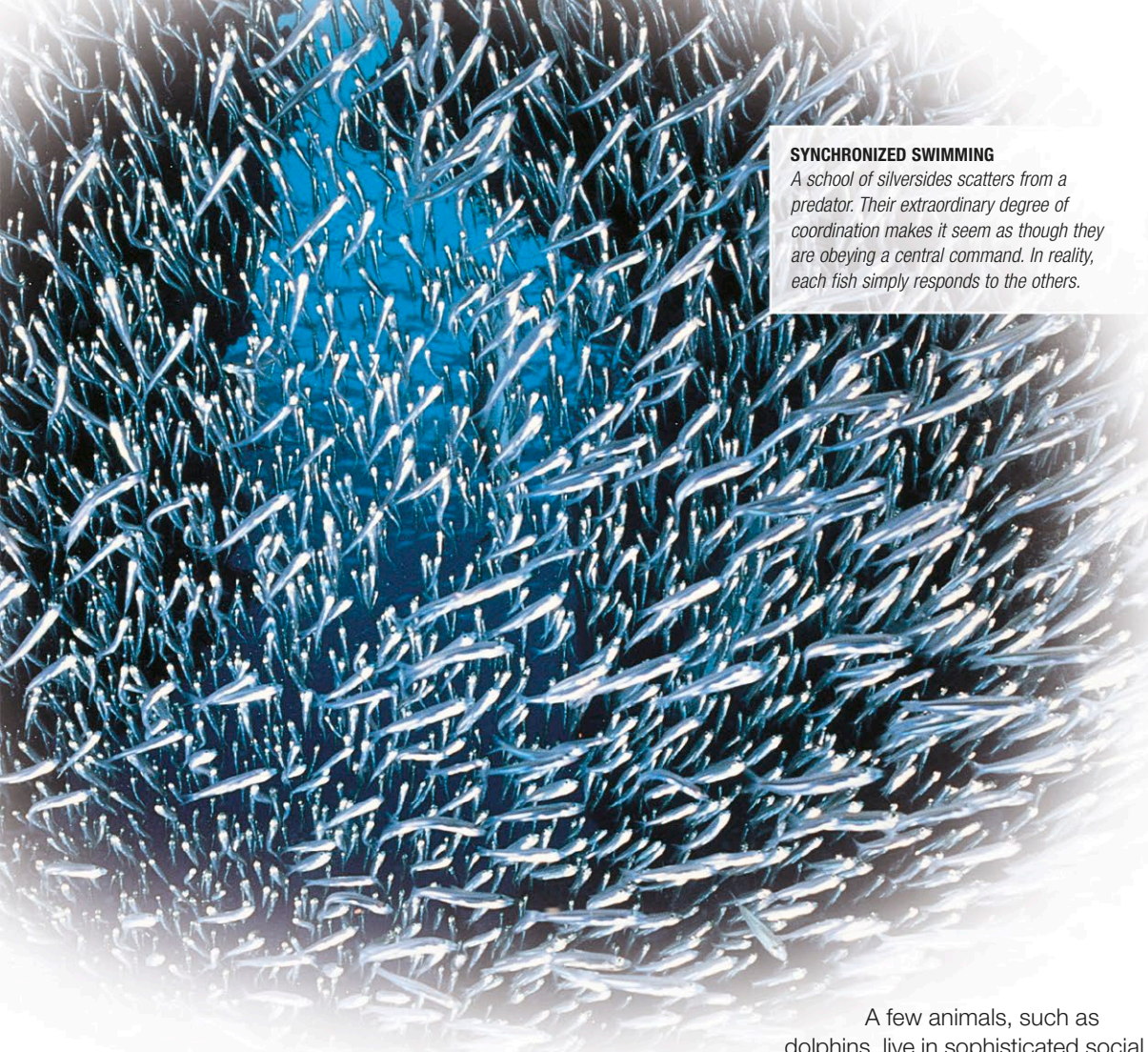


SYNCHRONIZED SWIMMING

A school of silversides scatters from a predator. Their extraordinary degree of coordination makes it seem as though they are obeying a central command. In reality, each fish simply responds to the others.

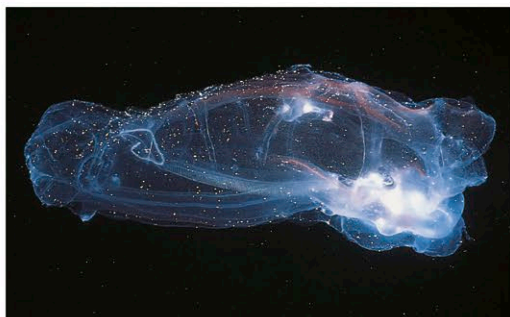


bright sky, and it conceals them from surface hunters, including seabirds, by making them blend in with the dark water beneath.

A few animals, such as dolphins, live in sophisticated social groups. Dolphins use echolocation to locate prey, to warn each other of danger, and to organize themselves during hunting.

Migration

Many marine animals—including most of the largest whales—migrate between breeding grounds in the tropics and feeding grounds at higher latitudes. The gray whale probably travels the greatest distances: its lifetime annual migrations total up to 500,000 miles (800,000 km)—twice the distance to the moon. Tuna are also known to undertake immense journeys: some schools travel the length of the Mediterranean; in the Pacific, one fish tagged in Mexico was recovered off Japan. Turtles show very precise migration patterns, returning to the same stretch of beach year

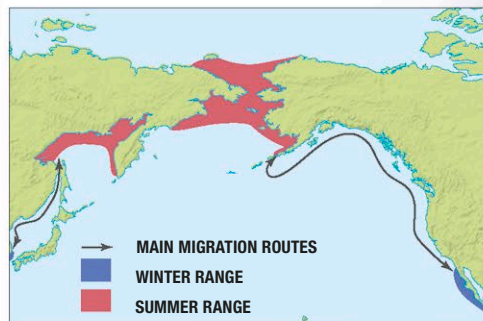
**TRANSPARENT LIFE**

Barrel-shaped salps have clear, jellylike bodies that are difficult for predators to see. Up to 4 in (10 cm) long, they live in swarms that can extend hundreds of miles across the ocean's surface.

Living in groups

On land, animal groups can be very large, but none rivals the size of those that can occur at sea. Fish often live in schools thousands or even millions strong, while some planktonic animals form swarms that can be over 60 miles (100 km) long.

These giant aggregations of underwater life often seem easy targets for predators. Whales gorge on krill with almost nonchalant ease, while other predators make huge inroads into schools of fish. But, in general, animals living in groups are safer than they would be alone: they are more difficult to single out and much more difficult to take by surprise.

**WHALES ON THE MOVE**

North American gray whales make an annual return trip between their winter breeding grounds off northwest Mexico and their summer feeding grounds around the Bering Sea. A smaller number migrate between Korea and eastern Siberia.

HYDROTHERMAL VENTS

First seen in 1977, off the Galapagos Islands, hydrothermal vents are remarkable ecological oases on the deep seabed. Vents are created by volcanic activity releasing streams of intensely hot, mineral-laden water into the ocean. They are among the few habitats where life does not rely on energy from the sun—specialized bacteria use the minerals to produce energy, and vent animals either consume food created by the bacteria or eat each other. Hundreds of animal species have been discovered around vents; most of these are not found elsewhere.

**GHOSTLY ANIMALS**

Like most vent animals, these crabs and clams—caught in the powerful lights of a submersible—have very little pigmentation, which explains their white appearance.

after year to lay their eggs. This is all the more remarkable because it takes 2 or 3 decades for the animal to mature: during this time, a turtle remembers precisely where it hatched so that when it is ready to breed it can make the long journey back.

On a smaller scale, many fish swim inshore to spawn, while other slow-moving animals migrate across the seabed. Caribbean spiny lobsters travel between shallow reefs, where they breed, and deeper water, where they overwinter; they set off in single file, each one following the tail of the animal in front.

Journeys like these are usually annual events. But some animals migrate daily to feed. Planktonic animals often rise to the surface at night, sinking back into the depths by day. Some of the plankton's predators copy this pattern, creating a 24-hour cycle involving many animals. These vertical migrations are clearly revealed by shipboard sonar, which shows a reflective layer rising at sunset and sinking at dawn.

